seth_dice_4.2

Seth Dice

2025-06-29

Remove any data that is not relevant to the patient's ALS condition.

Displaying column names from dataset to review which columns are not relevant to a patient's ALS condition.

```
##  [1] "ID"                               "Age_mean"
##  [3] "Albumin_max"                      "Albumin_median"
##  [5] "Albumin_min"                      "Albumin_range"
##  [7] "ALSFRS_slope"                     "ALSFRS_Total_max"
##  [9] "ALSFRS_Total_median"              "ALSFRS_Total_min"
## [11] "ALSFRS_Total_range"               "ALT.SGPT._max"
## [13] "ALT.SGPT._median"                 "ALT.SGPT._min"
## [15] "ALT.SGPT._range"                  "AST.SGOT._max"
## [17] "AST.SGOT._median"                 "AST.SGOT._min"
## [19] "AST.SGOT._range"                  "Bicarbonate_max"
## [21] "Bicarbonate_median"               "Bicarbonate_min"
## [23] "Bicarbonate_range"                "Blood.Urea.Nitrogen..BUN._max"
## [25] "Blood.Urea.Nitrogen..BUN._median" "Blood.Urea.Nitrogen..BUN._min"
## [27] "Blood.Urea.Nitrogen..BUN._range"  "bp_diastolic_max"
## [29] "bp_diastolic_median"              "bp_diastolic_min"
## [31] "bp_diastolic_range"               "bp_systolic_max"
## [33] "bp_systolic_median"               "bp_systolic_min"
## [35] "bp_systolic_range"                "Calcium_max"
## [37] "Calcium_median"                   "Calcium_min"
## [39] "Calcium_range"                    "Chloride_max"
## [41] "Chloride_median"                  "Chloride_min"
## [43] "Chloride_range"                   "Creatinine_max"
## [45] "Creatinine_median"                "Creatinine_min"
## [47] "Creatinine_range"                 "Gender_mean"
## [49] "Glucose_max"                      "Glucose_median"
## [51] "Glucose_min"                      "Glucose_range"
## [53] "hands_max"                        "hands_median"
## [55] "hands_min"                        "hands_range"
## [57] "Hematocrit_max"                   "Hematocrit_median"
## [59] "Hematocrit_min"                   "Hematocrit_range"
## [61] "Hemoglobin_max"                   "Hemoglobin_median"
## [63] "Hemoglobin_min"                   "Hemoglobin_range"
## [65] "leg_max"                          "leg_median"
## [67] "leg_min"                          "leg_range"
## [69] "mouth_max"                        "mouth_median"
```

```
##  [71] "mouth_min"                          "mouth_range"
##  [73] "onset_delta_mean"                   "onset_site_mean"
##  [75] "Platelets_max"                      "Platelets_median"
##  [77] "Platelets_min"                      "Potassium_max"
##  [79] "Potassium_median"                   "Potassium_min"
##  [81] "Potassium_range"                    "pulse_max"
##  [83] "pulse_median"                       "pulse_min"
##  [85] "pulse_range"                        "respiratory_max"
##  [87] "respiratory_median"                 "respiratory_min"
##  [89] "respiratory_range"                  "Sodium_max"
##  [91] "Sodium_median"                      "Sodium_min"
##  [93] "Sodium_range"                       "SubjectID"
##  [95] "trunk_max"                          "trunk_median"
##  [97] "trunk_min"                          "trunk_range"
##  [99] "Urine.Ph_max"                       "Urine.Ph_median"
## [101] "Urine.Ph_min"
```

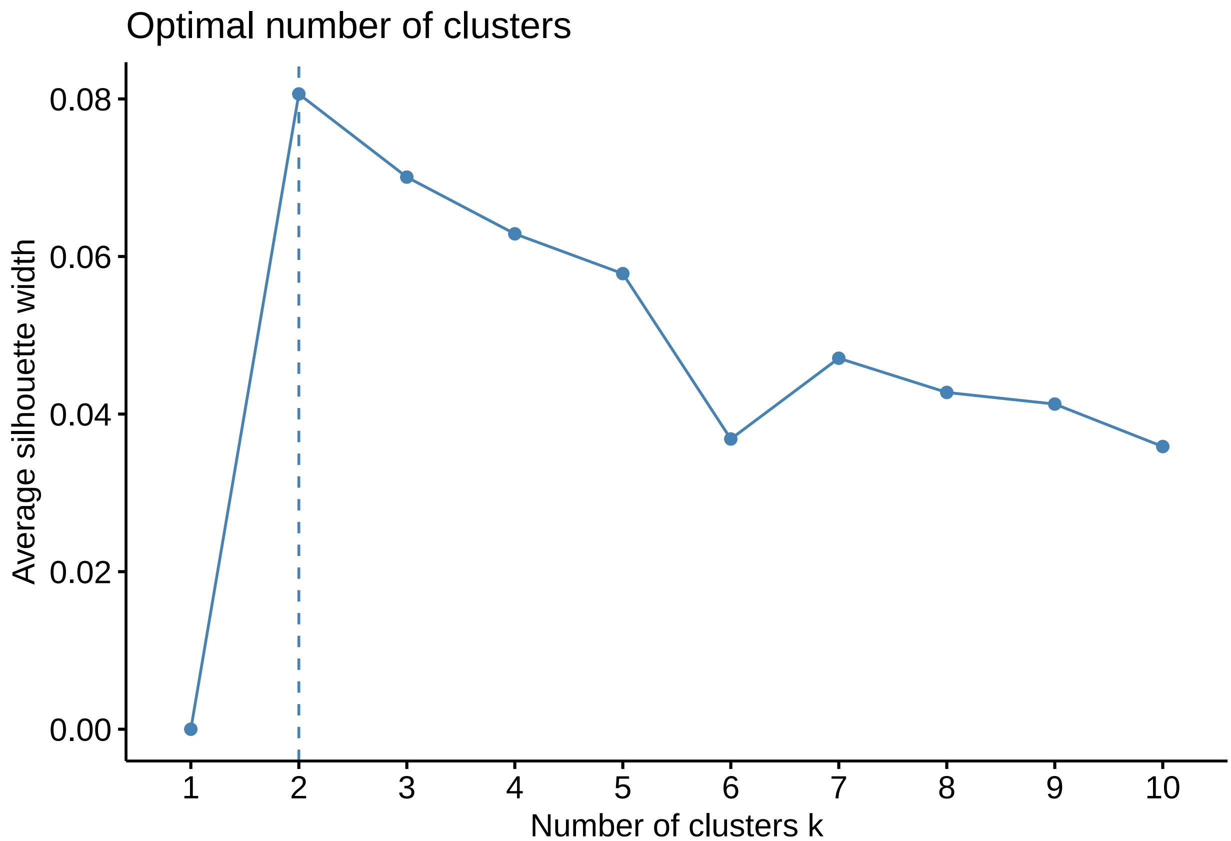
The columns that are deemed not important to ALS condition per Google resources:

- ID - Column ID given to patient
- SubjectID - Column ID given to subject
- Age_mean - Age is not relevant factor in ALS condition
- Gender_mean - Gender is not relevant factor in ALS condition

Apply a standard scalar to the data.

Used R-function scale() to apply standard scalar to ALS data.

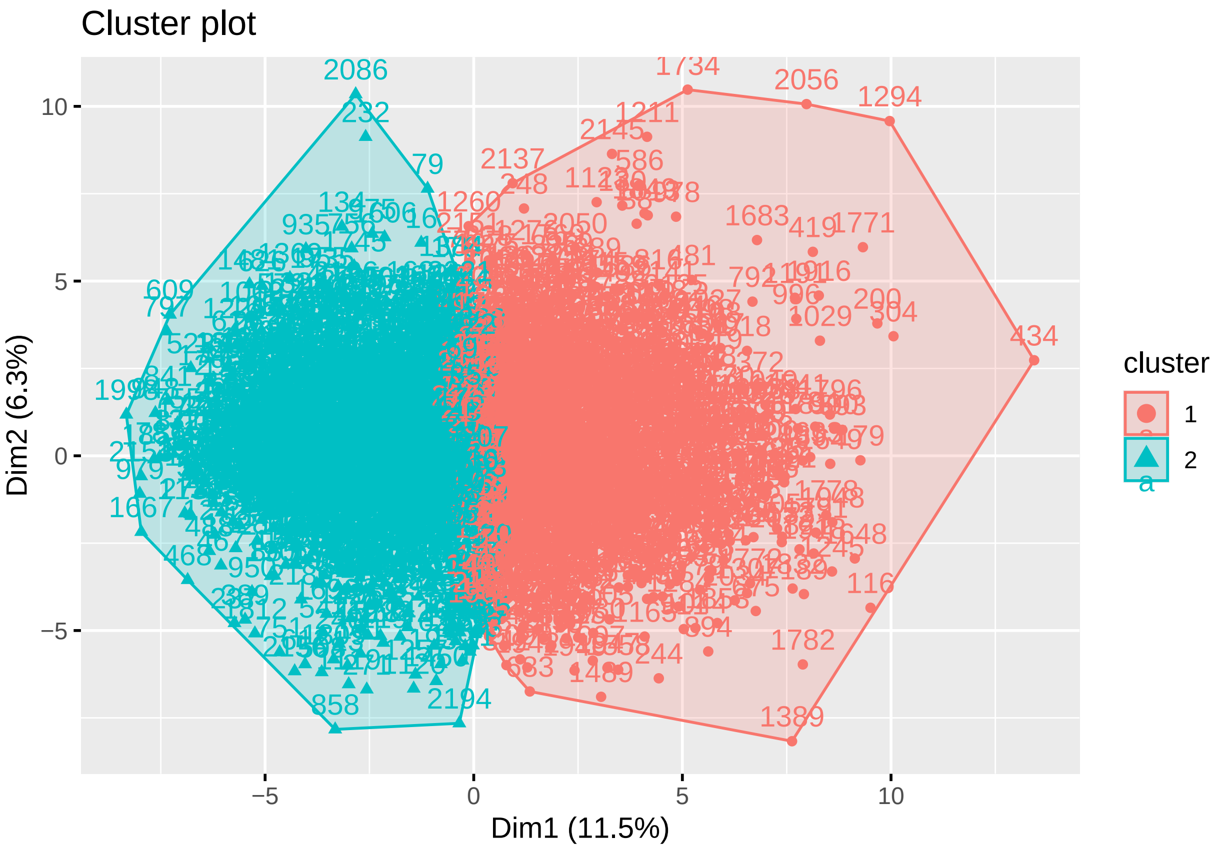
Create a plot of the cluster silhouette score versus the number of clusters in a K-means cluster.



Use the plot created in (3) to choose an optimal number of clusters for K-means. Justify your choice.

The Average Silhouette Method determines how well each value lies within its cluster. High average silhouette width indicates a good clustering, so from plot created in (3) we'd want the number of clusters with the highest average silhouette width. The highest average silhouette width is where number of clusters = 2.

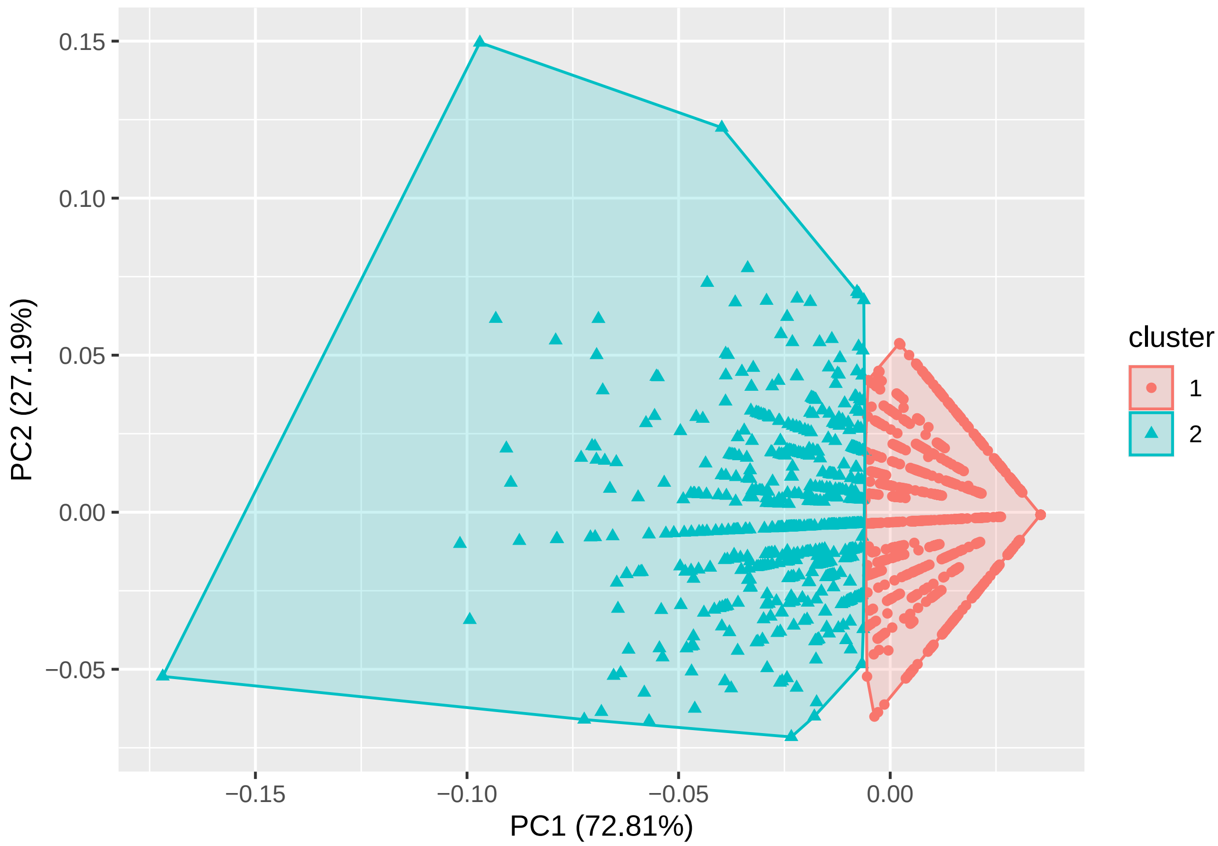
Fit a K-means model to the data with the optimal number of clusters chosen in part (4).



Fit a PCA transformation with two features to the scaled data.

The two features used from ALS data are: leg_range, hands_range

Make a scatterplot of the PCA transformed data coloring each point by its cluster value.



Summarize your results and make a conclusion.

After importing the ALS data and reviewing the columns the features removed were ID, SubjectID, Age_mean, and Gender_mean as Google research was not showing these as relevant to a patient's ALS condition. Once columns removed and scalar applied the optimal number of clusters selected was 2 when using the average silhouette method. A K-means model was then fitted with k = 2, and a PCA transformation was applied using two features chosen at random: leg_range and hand_range which is shown as a scatterplot. The K-means clustering graph shows the first cluster having a wider range of values when compared to the second cluster. The PCA scatterplot has vectors pointing in the same direction which would give a positive correlation to leg_range and hands_range.